

v1.0W Z Boson Deep Dive Simulation

V1.0 to V2.0 Versioning Delta — Tardigradia SubParticles Engine

Architecture: Unified Weak Field Architecture **Type:** Scientific Change Log | **Date:** June 2026 **Applies to:** v1.0W Z Boson Deep Dive Simulation V2.0.md

Revision Necessity — Four Drivers

1. **Experimental discoveries 2022-2026** — LHC Run 3, LIGO O4, KATRIN 2024, NIF ignition, and other landmark results supersede V1.0 physics parameters
2. **New confirmed particles and phenomena** — Pines’ Demon (2023), HH production evidence (2024), IceCube tau neutrino (2023), and others add entirely new simulation modes
3. **Platform upgrade** — WebGPU (Chrome 113+ 2023) replaces WebGL; enables 10^{7+} particles
4. **Precision improvements** — CODATA 2022 constants, FLAG 2024 lattice QCD, NuFIT 5.3 oscillations

V1.0 vs V2.0 Specific Changes

Parameter	V1.0	V2.0	Severity	Consequence if Not Updated
W boson mass anomaly	CDF anomaly as confirmed	CMS 2024: 80.360 GeV (SM-consistent). ATLAS 2023: 80.360 GeV. CDF anomaly NOT confirmed by other experiments	HIGH	V1.0 may have treated CDF anomaly as confirmed physics — status changed to ‘unresolved’
Z width N _ν	N _ν mentioned	Gfitter 2024: N _ν = 2.9963±0.0074. Definitively 3 active neutrino generations. Z_WIDTH precisely updated	LOW	Precision constraint on neutrino generation count
EW precision S,T,U	V1.0 era fits	Gfitter 2024 with CMS W-mass: S=0.02±0.12, T=0.05±0.12 — more SM-consistent than CDF scenario	MEDIUM	EW precision fit changes with W mass update — oblique parameter recalculation

Parameter	V1.0	V2.0	Severity	Consequence if Not Updated
WW scattering unitarity	Brief mention	ATLAS WW 2023: Higgs restores unitarity in $W_L W_L \rightarrow W_L W_L$ — TGC measured precisely	MEDIUM	Direct measurement of Higgs contribution to WW unitarity now available

Simulation Impact Summary

The changes above drive the following modifications to the game engine architecture:

- **New particle types:** W
 - **Parameter updates:** All values in `static constexpr` declarations refreshed from 2022-2026 data
 - **WebGPU migration:** Particle update loop moves from CPU JavaScript to WGSL compute shaders
 - **New interaction modes:** Triggered by new experimental discoveries
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Unchanged Architecture

All foundational philosophy is **carried forward unchanged**: - Worldline Monism (One-Particle Universe) ontology - Proper time τ as primary particle identifier - Intrinsic vs. Extrinsic property distinction - Benevolence metric (C·F·S product formula) - Object-Oriented data structure (struct ParticleInstance extends to V2.0)

End of Versioning Delta | v1.0W Z Boson Deep Dive Simulation V1.0 archived | V2.0 is the active specification as of June 2026